`Dynamic RAM (DRAM) Report

Name: Mohamed Khaled Qader

Faculty: Engineering

Department: Electronics and Communications Engineering

Instructor: Ms. Batoul

Date: 21/12/2025

Table of Contents

Introduction …… 2

How DRAM Works …… 2

Structure of DRAM …… 3

Advantages of DRAM …… 3

Disadvantages of DRAM …… 4

Applications of DRAM …… 4

Conclusion …… 5

References …… 5

1. Introduction

Dynamic Random Access Memory (DRAM) is one of the most common types of computer memory used in modern electronic devices. It is a type of volatile memory, meaning it loses its data when the power is turned off. DRAM is widely used in computers, smartphones, and other digital devices due to its relatively low cost and high density.

2. How DRAM Works

DRAM stores each bit of data in a tiny capacitor within an integrated circuit. These capacitors can either be charged (representing a “1”) or discharged (representing a “0”). Unlike Static RAM (SRAM), DRAM needs to be refreshed thousands of times per second because the charge in the capacitors gradually leaks away. This refreshing process allows DRAM to temporarily retain data while the system is powered on.

3. Structure of DRAM

A basic DRAM cell consists of two main components:

Capacitor: Stores the electrical charge representing data.

Transistor: Acts as a switch to control the reading and writing of data.

The cells are organized in a grid of rows and columns, allowing the memory controller to access each cell by specifying its row and column addresses.

4. Advantages of DRAM

High Density: Can store a large amount of data in a small physical space.

Low Cost: Cheaper to produce than SRAM, suitable for main memory.

Widely Available: Found in almost all modern computers and digital devices.

5. Disadvantages of DRAM

Volatility: Loses data when power is turned off.

Needs Refreshing: Constant refreshing slightly slows performance.

Slower than SRAM: Access times are longer compared to Static RAM.

6. Applications of DRAM

Personal Computers (PCs): Main system memory.

Laptops and Smartphones: Primary memory for apps and processes.

Graphics Cards: Sometimes used as frame buffer memory.

Embedded Systems: Used in devices requiring temporary high-density storage.

7. Conclusion

DRAM is a critical component of modern electronic systems due to its high storage capacity and cost-effectiveness. While it has some limitations, such as the need for constant refreshing and slower speed compared to SRAM, its advantages make it the most widely used type of volatile memory in computers and other digital devices. Understanding DRAM is essential for anyone studying electronics and communications engineering.

8. References

Hennessy, J. L., & Patterson, D. A. (2019). Computer Architecture: A Quantitative Approach.

Stallings, W. (2018). Computer Organization and Architecture.

Mano, M. M., & Kime, C. R. (2017). Logic and Computer Design Fundamentals.